HARNESSING THE POWER OF CNNs FOR AUTOMATED SKIN CANCER CLASSIFICATION

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Link to Google Colab

Introduction

Skin cancer is among the most common malignancies worldwide, with over 80,000 cases reported in Canada each year [1]. Identifying it at an early stage greatly improves patient outcomes, with five-year survival rates reaching up to 99% [2]. However, limited access to dermatologists—particularly in rural areas—creates barriers to timely diagnosis, and visual inspection alone yields only about 60% accuracy [3]. These challenges highlight the need for automated, reliable diagnostic tools. To address this gap, this project will apply deep learning, specifically convolutional neural networks (CNNs), to classify dermoscopic images from the HAM10000 dataset into seven skin lesion categories [4]. CNNs are well-suited for this task, as their layered architecture captures both low- and high-level features—such as color variation and border irregularities—that dermatologists rely on for diagnosis [5]. This project represents an initial step toward building deployable tools that can be integrated into clinical workflows to enhance the efficiency of the diagnostic process.

Illustration

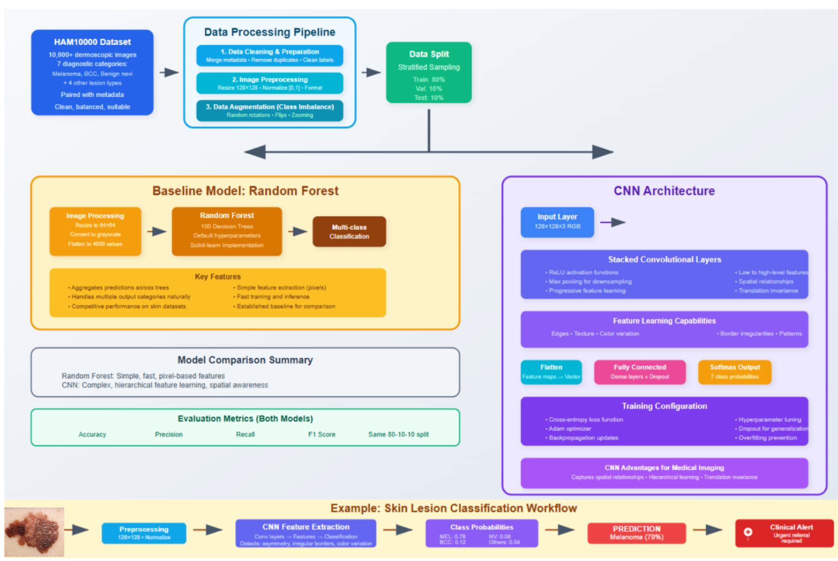

Figure 1: Proposed pipeline for skin lesion classification.

Background & Related Work

Deep learning has become central to skin lesion analysis, with numerous works showing dermatologist-level performance. A Skin Lesion Cancer feature-extractor CNN (SLC-CNN) was developed with preprocessing steps such as data augmentation, noise reduction, and normalization, using SVM for classification and XGBoost for segmentation. On the ISIC dataset, it achieved 99.6% classification and 95.3% segmentation accuracy, showing the value of hybrid pipelines with preprocessing [6]. Another work fused deep features from AlexNet, VGG16, and ResNet-18, with an SVM classifier on the ISIC 2017 dataset, reaching AUCs of 97.6% for seborrheic keratosis, demonstrating how complementary CNN representations can boost performance [7]. In addition, the Attention-Based Inception-Residual CNN (AIR-CNN) was introduced for multiclass skin cancer classification, integrating inception, residual, and attention blocks to better capture lesion features. On the ISIC 2019 dataset, it achieved 91.6% accuracy, showing how attention modules can improve feature focus and reduce errors among visually similar lesion types [8]. Further study fine-tuned pretrained CNNs (InceptionV3, DenseNet-201) and combined them in an ensemble for dermoscopic skin lesion classification. Test accuracy improved from 79.9% (Inceptionv3 with top-layer tuning) to 88.5% with a full fine-tuned ensemble, showing that well-tuned CNNs remain a strong baseline and warrant rigorous comparison [9]. Lastly, a custom FCDS-CNN outperformed pre-trained models (ResNet152V2, EfficientNetV2B0, InceptionResNetV2, MobileNetV3) with 96% accuracy, showing that a task-specific CNN can surpass generic models while being more resource-efficient [10].

Data Processing

This project will use the HAM10000 dataset, which contains over 10,000 dermoscopic images of pigmented skin lesions across seven diagnostic categories, including melanoma, basal cell carcinoma, and benign nevi. Each image is paired with metadata and diagnostic labels. Preprocessing will begin by merging and cleaning the metadata, removing duplicate lesion IDs to prevent data leakage, and discarding images with missing or ambiguous labels. All images will be resized to 128×128 pixels, converted to a consistent color format, and normalized to the [0,1] range. Labels will be standardized and encoded into seven classes. Since the dataset is highly imbalanced, with classes such as melanocytic nevi dominating while others are underrepresented, augmentation techniques (e.g., random rotations, flips, and zooming) will be applied to expand minority classes and improve generalization. The dataset will then be split into training, validation, and test sets in an 80-10-10 ratio using stratified sampling to preserve class proportions. These steps ensure that the dataset is clean, balanced, and suitable for training the CNN.

Architecture

The model architecture will be a CNN for classifying dermoscopic images into seven lesion categories. Input images will be resized and normalized before passing through stacked convolutional layers with ReLU activation and max pooling, allowing the network to progressively learn both low-level and high-level features such as edges, texture, color variation, and border irregularities. The resulting feature maps will be flattened and passed through fully connected layers, with dropout applied to improve generalization and reduce overfitting. A softmax output layer will then generate class probabilities for the seven lesion types. Training will use cross-entropy loss and the Adam optimizer, with parameters updated via backpropagation. Hyperparameters such as learning rate, batch size, and number of epochs will be tuned to optimize performance.

Baseline Model

The baseline model will be a Random Forest classifier trained on flattened grayscale images. Each image will be resized to 64×64 pixels, converted to grayscale, and flattened into a one-dimensional feature vector of 4096 values. The `RandomForestClassifier` from scikit-learn will be used with 100 decision trees and default hyperparameters. Training and evaluation will follow the same 80-10-10 train-validation-test split as the CNN, and performance will be measured using accuracy, precision, recall, and F1 score. Random Forests are well-suited for multiclass classification tasks, as they aggregate predictions across trees to handle multiple output categories, and prior work has shown they can achieve a competitive accuracy of nearly 95% on the HAM10000 dataset [11].

Ethical Considerations

Skin cancer classification raises significant ethical concerns, particularly regarding fairness and the consequences of misclassification. A key issue is that commonly used datasets like HAM10000 contain very few examples from patients with darker skin, reducing skin tone diversity and increasing the risk of poor accuracy for underrepresented groups [12]. Misclassification also carries serious impacts: false negatives may delay treatment, while false positives could lead to unnecessary procedures. These challenges underscore the need to assess model performance across diverse populations and to communicate limitations clearly so results are interpreted responsibly.

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